

**7 (short)** In a half-adder circuit, two single bit numbers  $X$  and  $Y$  are added and the result is stored in a two-bit number  $Z_1Z_0$ , where  $Z_0$  represents the least significant bit.

- (a) Using a truth table, derive the logic expressions for  $Z_1$  and  $Z_0$ . [4]
- (b) If any two-input logic gate can be used, what is the minimum number of gates required to implement the half-adder? What gates should be used? [2]
- (c) Implement the half-adder using NAND gates only. [4]

**8 (short)** A 256 kilobyte memory chip has 16 data lines.

- (a) How many address lines does the memory chip have? [4]
- (b) In a microprocessor with 20 address lines, how many of these memory chips can be connected? What are the address ranges of these chips in hexadecimal format? [6]